MILITARY SPECIFICATION

BOOSTER, M120
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of booster designated as Booster, M120.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

D

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8835171 — Booster, M120, Assembly.

8835190 — Carton, Packing, Ammunition, for Booster, M120 and Booster Retainer.

8835181 — Box, Packing, Ammunition, for Booster, M120 and Booster Retainer.

8837258 — Carton, Packing, Ammunition, for Booster, M120

8837259 — Box, Packing, Ammunition, for Booster, M120.

PUBLICATIONS

ORDNANCE CORPS

IEL-8835171 — Index of Inspection Equipment Lists.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Booster assembly. The booster assembly shall comply with all requirements specified on Drawing (dwg.) 8835171 and with all requirements specified in applicable specifications.

3.3 Waterproofness and functioning. The booster assembly shall function with high order detonation as evidenced by the absence of undetonated explosive and the destruction of an aluminum cylinder, 1½ inch long by 1⅛ inch diameter, when subjected to the waterproofness and functioning tests as specified in 4.4.2.

3.4 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

4.4.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Statements of findings on all composition, powder and associated ingredients, and certificates of conformance on all remaining material when such material is covered by Government specifications.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agents's

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authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2. *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production. During initial production, the provisions of 4.3 shall apply and all the examinations and tests therein shall be performed, except that the following test shall be conducted on the special sample quantities specified beginning with the first lots produced and continuing until three (3) consecutive lots have passed the acceptance criteria prescribed. Thereafter, sampling plans for this test shall be in accordance with 4.3.3.2.

4.2.1 *Waterproofness and functioning (see 3.3).* A sample of 80 booster assemblies shall be selected from each lot for this test. If three or more booster assemblies fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.4.4.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of boosters loaded and assembled by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with MIL–STD–105 (or MIL–STD–1235 when applicable). Each lot shall contain:

(a) Parts of one lot interfix number from one manufacturer.

(b) RDX from not more than one lot.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL–STD–1235 may be used, for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL or 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Pellet, booster (see dwg. 8835174 covering a detail of dwg. 8835171).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight, minimum (min.)	Balance	01001
Minor:	AQL 1.00 percent		
201.	Pellet cracked, chipped or broken	Visual	01002
202.	Evidence of poor workmanship (see 3.4)	Visual	01003

4.3.2.2. Assembly (see dwg. 8835171).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Explosive exposed or on exterior	Visual	02001
Major:	AQL 0.65 percent		
101.	Total length, maximum (max.)	Gage	02002
102.	Diameter	Gage	02003
103.	Assembly damaged (distorted, dented, deep cut, or wrinkled)	Visual	02004
104.	Crimp missing, improper, or less than 360 degrees	Visual	02005
Minor:	AQL 1.50 percent		
201.	Burrs or raised metal	Visual	02006
202.	Marking misleading or unidentifiable	Visual	02007
203.	Crack at crimp	Visual	02008
204.	Evidence of poor workmanship (see 3.4)	Visual	02009

4.3.2.3 Carton, sealed (see dwg. 8835190 or 8837258, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual	03001
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least 1/2 inch around entire carton	Visual	03002

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Categories	*Defects*	*Method of inspection*	*Code No.*
Minor:	AQL 1.00 percent		
201.	Contents move when carton is shaken	Manual	03003
202.	Marking misleading or unidentifiable	Visual	03004

4.3.2.4 Bag, sealed (see dwg. 8835190 or 8837258, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Waterproofness destroyed by separation of heat seal, or rip, tear, or puncture	Visual	04001
Minor:	None defined		

4.3.2.5 Box, sealed wooden (see dwg. 8835191 or 8837259, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Hardware missing, broken or loose	Visual-Manual	05001
102.	Strapping missing, broken or loose	Visual-Manual	05002
103.	Contents exposed	Visual	05003
104.	Board broken or split	Visual	05004
105.	DOD symbol misleading or unidentifiable	Visual	05005
Minor:	AQL 2.50 percent		
201.	Hardware improperly engaged	Visual-Manual	05006
202.	Handle missing or insecure	Visual-Manual	05007
203.	Contents loose (when box is tilted)	Visual	05008
204.	Marking misleading or unidentifiable	Visual	05009
205.	Strapping improperly assembled	Visual	05010
206.	Car seal missing, unsealed, or improperly positioned	Visual	05011

4.3.3 *Testing.*

4.3.3.1 *Determination of total binder-lubricant content of RDX booster pellets (see dwg. 8835174 covering a detail of dwg. 8835171) — Major defect — Code No. 01004.* Ten (10) RDX booster pellets shall be selected from each lot, from the loading line, prior to assembly into the cup. The

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tests shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2. *Waterproofness and functioning (see 3.3) — Major defect — Code No. 02010.* The sampling plan for this test shall be in accordance with MIL–STD–105 using Code letter H and an AQL of 1.0 percent. Any booster assembly which fails to comply with the applicable requirements shall be classed defective. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8835171 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of it's accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Determination of total binder-lubricant content of RDX booster pellets.* The percentage of total binder-lubricant shall be determined by adding the percentages of stearic acid and graphite found as specified in 4.4.1.1. and 4.4.1.2, respectively.

4.4.1.1 *Stearic acid content.* The outer surface of the booster pellets shall be removed and a composite sample of about 5 grams (gms.) of RDX shall be obtained from the interior of the booster pellets. An accurately weighed portion of 5 grams of the sample shall be transferred to a tared, medium porosity, sintered-glass filtering crucible. The sample shall be extracted with three 20-milliliter (ml.) portions of chloroform which has been previously saturated with RDX, each portion being maintained in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up by occasional stirring and the sides of the crucible shall be washed down with each 20-ml portion of chloroform to insure complete removal of the stearic acid. The residue shall be aspirated until the odor of chloroform is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees centigrade to 105 degrees centigrade for ½ hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedures shall be repeated with individual 20-ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 gm. The loss in weight of the crucible, plus contents of percent stearic acid in the sample, shall be calculated on a moisture-free basis as follows:

$$\text{Percentage of stearic acid} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible plus contents.

W = weight of sample on a moisture-free basis.

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4.4.1.2 *Graphite content.* The residue remaining after the stearic acid has been removed shall be extracted with two 20-ml portions of dimethyl formamide, maintaining each portion in contact with the residue for approximately 5 minutes, with intermittent stirring, before applying suction. The crucible and contents shall be washed with a minimum of three 20-ml portions of acetone, maintaining each portion in contact with the residue for approximately 1 minute, with continuous stirring before applying suction. This action shall be continued until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25-ml beaker, evaporated to dryness on a steam bath, and several drops of a solution of chromotropic acid in 36N sulfuric acid added. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner, indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees to 105 degrees centigrade for ½ hour, cooled in a desiccator, and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture-free basis as follows:

$$\text{Percentage of graphite} = \frac{100A}{W}$$

where:

A = increase in weight of filtering crucible.

B = weight of sample on moisture-free basis.

4.4.2 *Waterproofness and functioning.* The booster assemblies shall be completely immersed in water for a depth of not less than 8 inches for a period of 24 to 25 hours. The water shall contain a wetting agent such as naccatol. Within one-half hour after removal from the water, the booster shall be assembled to equipment in accordance with 4.3.4 and functioned to determine compliance with the applicable requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Booster, M120, shall be packaged in accordance with dwg. 8835190.

5.1.2 *Level C.* Booster, M120, shall be packaged in accordance with dwg. 8837258.

5.2 Packing.

5.2.1 *Level A.* Booster, M120, shall be packed in accordance with dwg. 8835191.

5.2.2 *Level C.* Booster, M120, shall be packed in accordance with dwg. 8837259.

5.3 Marking. Marking shall be as specified on the applicable drawing.

5.4 Data cards. Data card information shall be prepared for each lot in accordance with MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability

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of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF	— Contractor's design responsibility on Army format in accordance with MIL–D–45608.
CDCF	— Contractor's design responsibility on contractor format.
AD	— Army design.
ADMU	— Army design, mandatory for use.
CE	— Commercial equipment.
SCD	— Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

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6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic full size prints of top quality with black lines on white background, and shall provide a legible print produced via the 35mm microfilm process. The contractor shall submit one rolled or flat set copy of the final approved design.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1345–A164

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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